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Question NO.1: Answer briefly each part.

[3X10]

i. Assume following is loaded starting at address 0100h.

A DW 1111, 2222, 3333, 4444, 5555, 6666, 7777, 8888
 B DW 1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000
 C DB 8 DUP (34H)
 D DW 1234, 2345, 3456, 4567, 5678, 6789, 7788, 8899

Give the offset address assigned to value 5678. What value is stored at address 0122h?

0130
 ↓
 offset 5678

0122 $\Rightarrow$ 34h

ii. What will be value of SI and DI when the program will terminate assuming starting offset address assigned to array1 is 0150h?

MOS SI, OFFSET ARRAY1
 MOV DI, OFFSET ARRAY2
 MOV CX, 8
 AA: MOV BL, [SI]
 MOV [DI], BL
 INC SI
 INC DI
 DEC CX
 JNZ AA
 MOV AH, 4CH
 INT 21H

ARRAY1 DB 41H, 42H, 43H, 44H, 45H, 46H, 47H, 48H
 ARRAY2 DB 41H, 42H, 43H, 44H, 65H, 46H, 47H, 48H

	SI	0150	
	DI	0158	
0150		0158	8
0151		0159	7
52		015A	6
53		015B	5
54		5C	4
55		5D	3
56		5E	2
57		5F	1
58		60	0

SI = 0058
 0158

DI = 0060
 0160

iii. How many times the loop will be executed in the following code?

AA: MOV CX, 0
 MOV DL, 'A'
 MOV AH, 02
 INT 21H
 DEC CX
 JNZ AA

0-1
 FFFF

loop will be executed 10000 (FFFF + 1) times

viii.

Consider the following Memory Area shown by the dump (d) instruction. With reference to the data shown in part below, what will be the values of DH register after the execution of the instruction ADD DH, [01FE] where DH = 40h?

0C21:01A0	13 96 D0 E0 D0 E0 A2 1E - 99 80 3E 20 99 00 75 24	> ..u\$
0C21:01B0	A2 24 99 0A C9 75 1D 0A - C0 74 19 8B 0E 21 F7 D8	\$.u...t...l..
0C21:01C0	13 80 1A 06 33 FF 8E 06 - 00 96 F2 AE 07 75 05 4F	3.....u.O
0C21:01D0	89 3E 21 96 8B 06 97 80 - 3E 13 96 00 74 03 8B 4C	>>...t..L
0C21:01E0	37 8E C3 96 8B 3E 05 98 - 89 08 00 AC 3C 3F 75 02	>.....<?u.
0C21:01F0	8A 07 3C 20 74 01 AA 43 - E2 F1 B1 03 80 20 38 04	..< t..C..... 8.
0C21:0200	74 12 80 2E AA AC 3C 3F 75 02 8A 07 3C 20 74 01	t.....<?u...< t.
0C21:0210	AA 43 E2 F1 32 C0 AA C3 F6 46 04 02 75 43 8B A3	.C..2....F...uC..

[01FE] = '08'

ADD DH, [01FE] ; 40 + 38

DH = 78h

ix.

What will be in STRING when the program completes?

```

MOV SI, OFFSET STRING
AA:  MOV BL, [SI]
      CMP BL, 'S'
      JE EXIT
      AND BL, 0DFH
      MOV [SI], BL
      INC SI
      JMP AA
EXIT:

```

BL
G

0100 0111
0000 1101 1111

EXIT:

STRING DB 'Good Morning\$'

GOOD MORNING\$

'good morning\$'

x.

Assume AL=16h; BL=25h. What will be the status of Sign and Zero flags after the execution of instruction SUB AL, BL? Also mention the value stored in AL in hexadecimal.

ZF = 0 (Reset)

SF = 1

AL = 11110001

16h - 25h
 0001 0110
 - 0010 0101
 1111 0001

Question NO.2: Read the following program and answer what is stored in AX, BX, CX and DX in hexadecimal at the end of each iteration? [10]

```

MOV AX,0024H
MOV BX,0010H
MOV CX,0
MOV DX,0
AGAIN: TEST BX,1
JZ AA
ADD DX, AX
AA: INC CX
SHL AX,1
SHR BX,1
JNZ AGAIN
MOV AH,4CH
INT 21H

```

0000 0000 0010 0100
0100 1000

0000 0000 0001 0000

0000 0001 0010 0000

Iteration #	AX 0024	BX 0010	CX 0	DX 0
After 1 st iteration	0048	0008	1	0
After 2 nd iteration	0090	0004	2	0
After 3 rd iteration	0120	0002	3	0
After 4 th iteration	0240	0001	4	0
After 5 th iteration	0480	0000	5	0240

AX 0024 BX 0010 CX 0 DX 0

Question No.3:

The program hello.asm is shown in the left window and its exe file opened in debugger is shown in the middle window. Go through the program and answer the questions given at the end. Assume initially SP=0200h and AX=124Ah before the execution of the first instruction for all questions asked. Draw the stack in the right window. [10]

```
main proc
    mov ah,01
    int 21h
    mov dl,a1
    call display
    mov ah,4ch
    int 21h
main endp
```

```
display proc
    push ax
    push bx
    int 21h
    call xyz
    pop bx
    pop ax
    ret
display endp
```

```
xyz proc
    push cx
    mov dl,0Ah
    mov ah,02
    int 21h
    mov dl,0Dh
    mov ah,02
    int 21h
    pop cx
    ret
xyz endp
```

```
D:\Masm>debug first.exe
-u 0000 0023
13D6:0100    MOV     AH,01
13D6:0102    INT     21
13D6:0104    MOV     DL,AL
13D6:0106    CALL    000D
13D6:0109    MOV     AH,4C
13D6:010B    INT     21
13D6:010D    PUSH    AX
13D6:010E    PUSH    BX
13D6:010F    INT     21
13D6:0111    CALL    0017
13D6:0114    POP     BX
13D6:0115    POP     AX
13D6:0116    RET
13D6:0117    PUSH    CX
13D6:0118    MOV     DL,0A
13D6:0119    MOV     AH,02
13D6:011B    INT     21
13D6:011D    MOV     DL,0D
13D6:011F    MOV     AH,02
13D6:????    INT     21
13D6:0123    POP     CX
13D6:0124    RET
```

Draw stack here

	01F6
	01F8
CX	01FA
0114	01FC
BX	01FE
AX	0200
0109	

	01F6
CX	01F8
0114	01FA
BX	01FC
AX	01FE
0109	0200

- What is the value of IP and SP after the execution of instruction "CALL xyz"?
IP = 0017
SP = 01FB
- What would be in SP after the execution of PUSH CX stored at 13D6:0117?
SP = 01F6
- What would be in SP after the execution of PUSH BX stored instruction?
SP = 01FA
- What is the value of IP and SP after the execution of instruction "RET" stored at 13D6:0124?
IP = 0114
SP = 01FA
- What would be in SP after the execution of RET stored at 13D6:0116?
SP = 0200

Question No.4: Declare and initialize an array of 8 bytes in the data segment named as array1. Write down the code to find out the largest element and store it in BL register. Consider the numbers as signed.

```
.model small
.stack 100h
.386
.data
    array1 DB 10h, 12h, 14h, 16h, 18h, 20h, 22h, 24h
.code
main proc
    mov AX, @data
    mov DS, AX
    mov CX, 07
    mov SI, offset array1
    mov BL, [SI]
    inc SI
aa: mov AL, [SI]

    cmp BL, AL
    jg exit
    mov BL, AL
exit: inc SI

    dec CX
    jnz aa

    mov AH, 4ch
    int 21h

main endp
end main
```



Question No.5: Write a program that will get two 8-bit numbers from the user in binary number system, add them and display the result in Hexadecimal as output. [12]

```

.model small
.stack 100h
.386
.code
main proc
    mov     BX,0
    call    Fbin
    mov     BH,BL
    call    Fbin
    add     BL,BH
    call    Fhex
    mov     AH,4ch
    int     21h
main endp
end main

```

```

Fbin proc
    mov     CX,8
again: mov     AH,01
    int     21h
    sub     AL,30h
    shl     BL,1
    or      BL,AL
    dec     CX
    jnz     again
    ret
Fbin endp

```

11.5

```

Fhex Proc
    mov     CX,02
aa:
    rol     BL,4
    mov     DL,BX
    and     DL,0Fh
    cmp     DL,09h
    ja      let1
    add     DL,30h
let1:
    add     DL,37h
    mov     AH,02
    int     21h
    dec     CX
    jnz     aa
    ret
Fhex endp

```



```

      1111 1111 0000 0000
      0000 0000 1111 1111
      -----
            010000 0000
      D      1      00
  
```

- iv. Assume AX=8000h and BX=FF00h. What will be in DX after the execution of the following instructions:

```

      CMP AX, BX
      JLE AA
      MOV DX, 60H
      JMP EXIT
AA:    MOV DX, 40H
EXIT:
  
```

AX < BX

-8, -4

```

      1000 0001 0000 0000
      0111 1110 1111 1111
      -----
      1111 1111 1111 1111
  
```

DX = 40H

- v. A memory location has physical address 93FC4h that corresponds to offset address 1FC4h. Calculate the segment address and the maximum possible physical address in this segment.

$$P.A = S.A \times 10h + O.A$$

$$93FC4 = S.A \times 10h + 1FC4$$

$$93FC4 - 1FC4 = S.A \times 10h$$

$$\frac{92000h}{10h} = \frac{S.A \times 10h}{10h}$$

$$S.A = 9200$$

```

      93FC4
      1FC4
      -----
      92000
  
```

?

- vi. Write down assembly code for the following decision structure:

IF (DL >= 'a') AND (DL <= 'z')

Display DL

DL < 'a' → F

DL > 'z' → F

CMP DL, 'a'

JL exit

CMP DL, 'z'

JG exit

MOV AH, 02

INT 21H

exit

- vii. Assume AL and BL contain ASCII codes of two characters. Print the character that comes first alphabetically.

CMP AL, BL

JG aa

MOV DL, AL

JMP exit

aa: MOV DL, BL

exit: MOV AH, 02

INT 21H

AL = 42

BL = 41